higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 4

(12041004)

18 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages and a formula sheet of 3 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use only blue or black pens.
6. Write neatly and legibly.

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QUESTION 1

1.1 An electrical network is a system of interconnected elements, including resistors, inductors, capacitors, voltage sources, current sources and switches.

Name FOUR classifications of electrical networks. (4)

1.2 Refer to the circuit diagram in FIGURE 1 below and calculate the following:

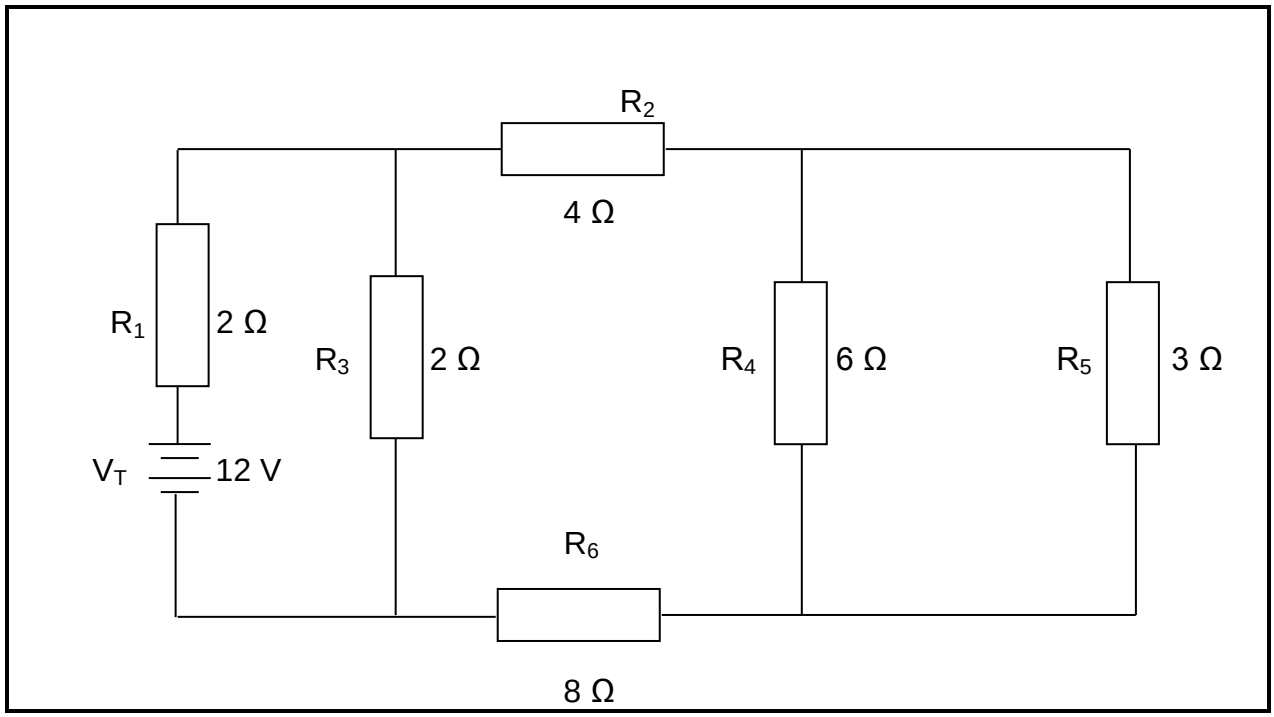


FIGURE 1

1.2.1 The total resistance of the circuit (7)

1.2.2 The current flowing through the R_1 resistor (2)

1.2.3 If resistor R_2 is disconnected from the circuit, determine the total resistance of the circuit. (2)

1.2.4 Determine the total current of the circuit as the result of disconnecting resistor R_2. (2)

1.3 The voltage applied to an AC circuit is presented by the equation $v = 200 \sin 628 t$.

Determine the following:

1.3.1 The rms value of the voltage (2)

1.3.2 The frequency of this voltage (2)

1.3.3 The instantaneous value of the voltage, 2 ms after the commencement of a cycle (3)

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1.4 Explain the difference between the *form factor* and the *peak factor* of a sine wave. (2)

1.5 The apparent input power to a single-phase motor is 7 kVA at a power factor of 0,6 lagging.

Determine the reactive power. (3)

1.6 What do you understand by the term *maximum value* of a sine wave? (1)
[30]

QUESTION 2

2.1 The generation of electricity is the conversion of other forms of energy into electrical current, one of type of resources that is mainly used to generate electricity is Hydro-electric energy.

Describe how hydro-electric energy is generated. (3)

2.2 State the SIX advantages of the three-phase distribution system. (6)

2.3 Star and delta connections are the two types of connections in three-phase circuits.

Compare these TWO connections with respect to their voltages. (2)

2.4 There are several components and materials used in overhead transmission lines.

Explain the purpose of poles in overhead transmission lines. (1)

2.5 A three-phase AC motor draws a current of 12 A when connected in delta to a 380 V supply at a power factor of 0,9.

Calculate the input power of the motor in kW. (3)
[15]

QUESTION 3

3.1 Lives can be saved if earthing is maintained and inspected regularly.

3.1.1 Give THREE reasons why it is important to earth the domestic installations and the electrical appliances (3)

3.1.2 Name THREE types of earth electrodes that can be used in the field of earthing. (3)

3.2 Describe how you would test the continuity of bonding in an electrical installation. (3)

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3.3 What do you understand by term *earth electrode*? (2)

3.4 What do you call the current that flows to earth in an electrically undamaged circuit? (1)

3.5 Explain what is meant by the following earthing systems:

3.5.1 TN-C-S (2)

3.5.2 TN-S (1)

[15]

QUESTION 4

4.1 Given a working step-down transformer without information(details), how would you identify the primary and the secondary terminals? (3)

4.2 A 300 kVA transformer operates at a power factor of 0,85.

Calculate the efficiency of a transformer at full load if the total losses amount to 1,9 kW. (5)

4.3 Your foreman has requested you to connect THREE single-phase transformers to produce ONE three-phase star/star transformer.

Indicate by means of a neatly labelled diagram how you would achieve this. (6)

4.4 The windings of a transformer are supported by a core.

Distinguish between hollow core and shell core type transformer. (2)

4.5 Complete the following table by writing only the answer next to the question number (4.5.1–4.5.4) in the ANSWER BOOK.

TRANSFORMER CONNECTIONS	APPLICATIONS
Delta-delta connection	4.5.1
Star-Star connection	4.5.2
Delta-star connection	4.5.3
Star-delta connection	4.5.4

(4 × 1) (4)

[20]

QUESTION 5

5.1 Separately excited generators are used in automatic motor control systems.

5.1.1 Make a neat fully labelled sketch of the characteristic curve of a separately excited generator. (4)

5.1.2 Explain the characteristic curve you have drawn in QUESTION 5.1.1. (4)

5.2 A three-phase induction motor is the type of electric motor that is used most extensively.

Explain the operation of a three-phase induction motor. (4)

5.3 A six-pole induction motor is connected to a 60 Hz supply. If the rotor runs at 900 rev/min,

Calculate the following:

5.3.1 The synchronous speed of the magnetic field (2)

5.3.2 The slip at full load (2)

5.4 Explain how the direction of rotation of a compound wound DC motor can be reversed. (2)

5.5 Induction motors need to be tested regularly to ensure they are working properly.

Explain the purpose of the open-circuit test on induction motor. (2)

[20]

TOTAL: 100

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ELECTRICAL PRINCIPLES AND PRACTICE L4

FORMULA SHEET

1. $\boldsymbol{Cos\phi = \frac{R}{Z}}$

2. $\boldsymbol{P = V\ I\ Cos\phi}$

3. $Q = V\ I\ Sin\phi$

4. $\boldsymbol{S = V\ I}$

5. $V_1 = \left(\frac{R_1}{R_t}\right) V_T$

6. $\boldsymbol{I_1 = \left(\frac{R_2}{R_1 + R_2}\right) I_T}$

7. $R_T = R_1 + R_2 + R_3 + \cdots + R_n$

8. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$

9. $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

10. $V_L = \sqrt{3}\, V\ p\ h$

11. $\boldsymbol{E = V - I_a\ R_a}$

12. $n_s = \frac{f}{p}$

13. $S = \frac{N_s - N_r}{N_s}$

14. $\boldsymbol{R_{sh} = \frac{I_m\ R_m}{I_{sh}}}$

15. $R_{se} = \frac{V_t}{I_t} - R_m$

16. $\tan\boldsymbol{\phi} = \sqrt{3}\left(\frac{\boldsymbol{W_1 - W_2}}{\boldsymbol{W_1 + W_2}}\right)$

17. $E = V + I\ r$

18. $\boldsymbol{V = I\ R}$

19. $R = \frac{p\ l}{A}$

20. $\boldsymbol{R_t = R_0(1 + \alpha t)}$

21. $R_t = R_0[1 + \propto_0 (t - \emptyset)]$

22. $P = V\ I$

23. $E = I^2\ R\ t$

24. $\boldsymbol{\eta = \frac{P_0}{P_{in}} \times 100}$

25. $\boldsymbol{X_L = 2\pi\ f\ L}$

26. $X_C = \frac{1}{2\pi\ f\ C}$

27. $\boldsymbol{Z = \sqrt{R^2 + (X_L - X_C)^2}}$

28. $Mmf = N\ I$

29. $H = \frac{Mmf}{l}$

30. $B = \frac{\Phi}{A}$

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31. $F = B\,I\,l$

32. $E = -L\dfrac{di}{dt}$

33. $A = \dfrac{\pi\, d^2}{4}$

34. $E = \dfrac{1}{2}\ C\,V^2$

35. $Q = C\,V$

36. $F = 2\times10^{-7}\times\dfrac{I_1\ I_2}{d}$

37. $I = \dfrac{Emf}{R + r_T}$

38. $L = N\dfrac{\Delta\,\phi}{\Delta\,I}$

39. $v = V_m \sin(\omega t\ \pm\ \emptyset)$

40. $Z = \dfrac{V}{I}$

41. $I_T = \sqrt{I_R^2 + (I_L - I_C)^2}$

42. $tan\emptyset = \dfrac{X_L - X_C}{R}$

43. $I_{sh} = \dfrac{V + I_{se}R_{se}}{R_{sh}}$

44. $T = \dfrac{0{,}318 I_a Zp\emptyset}{c}$

45. $E = V - I_a(R_a + R_{se})$

46. $E = B\,l\,v$

47. $E = -N\ \dfrac{d\emptyset}{dt}$

48. $E = \dfrac{V}{d}$

49. $E = \dfrac{1}{2}LI^2$

50. $Q = It$

51. $C_T = C_1 + C_2 + C_3 + \cdots + C_n$

52. $\dfrac{1}{C_T} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3} + \cdots + \dfrac{1}{C_n}$

53. $L_T = L_1 + L_2 + L_3 + \cdots + L_n$

54. $\dfrac{1}{L_T} = \dfrac{1}{L_1} + \dfrac{1}{L_2} + \dfrac{1}{L_3} + \cdots + \dfrac{1}{L_n}$

55. $tan\emptyset = \dfrac{I_L - I_C}{I_R}$

56. $E = V + I_a R_a$

57. $V = E - I(R_a + R_{se})$

58. $E = V + I_a(R_a + R_{se})$

59. $E = V + I_a R_a + IR_{se}$

60. $E = V - I_a R_a - IR_{se}$

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61. $E = \frac{2p\emptyset ZN}{60\ c}$

62. $V_T = \sqrt{V_R^2 + (V_L - V_C)^2}$

63. $V_L = IX_L$

64. $E_{rms}\ or\ I_{rms} = 0{,}707\ E_m\ or\ I_m$

65. $N = \frac{V}{k\emptyset}$

66. $E = 4{,}44\ f\emptyset_m N$

67. $V_C = IX_C$

68. $E_{ave} or\ I_{ave} = 0{,}637\ E_m\ or\ I_m$

69. $P = \sqrt{3}V_L I_L\ cos\theta$

70. $S = \sqrt{3}V_L I_L$

71. $Q = \sqrt{3}V_L I_L\ sin\theta$

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